Unsupervised Learning Laboratories

Lab 2

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2 Principal Component Analysis

In this lab we implement Principal Component Analysis (PCA), the first dimensionality reduction method of the course, and test it on a real dataset and on the Swiss Roll of Lab 1.

You can use external libraries for linear algebra operations (e.g. `numpy.linalg`) but you are expected to write your own PCA algorithm.

2.1 Exercise 1

Use the `Dry_Bean_Dataset.xlsx` available on the GitHub page of the labs.

- Divide your dataset into a train and a test set.
- Preprocess the data by centering the variables and dividing them by their standard deviation. Compute mean and standard deviation on the training set only, and use them also to transform the test set.
- Write your own algorithm to perform PCA on the variables.
- Using the training set, obtain and plot the eigenvalue spectrum using the log-scale for the y-axis. Plot also the cumulative fraction of explained variance. What number of principal components would you select?
- Project the data (training set) onto the first two principal components and color by class. Do it also for the first three principal components.
- For an increasing number of principal components (1 to 16):
  - Apply a multinomial logistic regression to learn a model on the training set (use `sklearn.linear_model.L`
  - Transform the test set with the matrix learned from the training set and make a prediction with the logistic model learned.
  - Assess the quality of the predictions and comment on the results.

Hint: The principal directions are the eigenvectors of the covariance matrix $C = \frac{1}{n-1} X^\top X$ of the centered data, ordered by decreasing eigenvalue.

2.2 Exercise 2

Apply PCA to the Swiss Roll dataset ($n = 1000$) from Lab 1 and plot the projection on the first two principal components. Color the points by the position ϕ along the roll and comment on your results: does PCA "unroll" the manifold?

Bonus

Compute the reconstruction error $\frac{1}{n}\sum_i \|x_i - \hat{x}_i\|^2$ on the test set as a function of the number of principal components, and compare it with the eigenvalues you found in Exercise 1.